Question 1

"Investing directly in real estate requires a good understanding of the specific nature of this business – and in particular of its specific risks and returns".

a- What are the main real-estate specific risks that make 'investing in buildings' directly different from investing in other financial assets? (10 points)

These issues were discussed at length in Lecture 1 and then again at the beginning of the lecture on REITs to provide a rationale for indirect investments in real estate. Risks include: operating expenses, vacancy, leasing, natural disaster, poor liquidity, large lot size, private information.

b- What are instead the main benefits of direct real estate investment, and what would you consider as most important for 'value generation'? (10 points)

These issues were discussed at length in Lecture 1 and include: operating costs, rental growth and resale value. The business of doing real estate is about generating good rental value. This affects sale price. Managerial practices that help – among other things – to reduce operating costs greatly help.

c- In what sense purchasing shares of Real Estate Investment Trusts (REITs) allows investors to gain exposure to real estate while by-passing some of these risks? (10 points)

These issues were discussed in Lecture 7. In a nutshell, REIT shares provide liquidity, information and can be purchased in small share – allowing for more diversified (and potentially less leveraged) portfolios of real estate indirect investments.

d- Explain the main characteristics of REITs – including constraints and possible expansion strategies – and discuss some commonly used valuation approaches. (20 points)

These issues were discussed at length during Lecture 7 both by the main lecturer and the guest speaker. The most important aspects to mention are: no-double taxation but constraints in terms of distribution of dividends. The most commonly used valuation approach was NAV – once again discussed by both the main lecturer and the guest speaker.

e- What is the relationship between REITs valuation and gearing over the cycle? And why some REITs trade at a premium and some at a loss? Provide answers to both points using the insights presented by the guest lecturer. (20 points)

These issues were brought to the attention of the students by the case speaker. They are well summarised in the set of slides he distributed and was made available on Moodle.

f- What are the main considerations you would keep in mind when structuring a portfolio of *indirect* real estate investments? (15 points)

g- What are the main considerations you would keep in mind when focussing on *direct* real estate investments? (15 points)

The issues discussed at both points f- and points g- were presented during Lecture 8 by Dr. Niko Szumilo who will guest-lecture on the course on these specific topics. The slides of the lectures summarise the main points discussed and that should be made by the students.

For resit candidates I will substitute f and g with:

f- Alternatively, investors can gain exposure to real estate by purchasing mortgage-backed securities. Discuss how the securitization process works and explain how different risks are apportioned to different financial actors by contrasting the structures of Mortgage Backed Bonds (MBB), Mortgage Pass-Through Securities (MPT) and Collateralized Mortgage Obligations (CMOs). (15 points)

These issues were discussed during Lecture 9. The main risks moved around by the process of securitization are default and (pre-)payment risks. MBB do not pass any of these risks to the investors – they are all held by the underwriter. MPT pass all of the risk although guarantees can be taken as most of these securities will have been some default insurance (vis-à-vis a fee). CMOs pass through risks but tranching alters the way in which different investors are exposed.

g- Consider now sub-prime mortgages. What are their main characteristics? And why are sub-prime mortgage-back securities more complex, risky and difficult to price than standard mortgage-backed securities? (15 points)

These issues were discussed in Lecture 10. Sub-prime mortgages are those extended to people who cannot afford prime contracts and thus have lower credit-worthiness. Subprime mortgages are usually ARMs with profitability (in the eyes of the bank) depending on positive house price dynamics. Securities of these mortgages are difficult to price because the overcollateralization is not automatic but depends on the performance of the mortgages. Moreover there are several triggers caused by delinquencies that make understanding the riskiness of these securities a complex task.

Question 2

You are considering purchasing a real estate investment worth £6 million. You would leverage this investment and the bank is offering a loan with an LTV of 50%, charging an annual interest of 6% and lasting for 30 years. Compounding is done monthly and the bank is offering three different types of loan: an interest only (IO) loan; a constant amortization loan (CAM) and a constant payment loan (CPM)

a- What are the payments due to the bank in the first month for these three alternative loans? And what are these payments at the end of the first year, at the end of the fifth year and in the last month of your contract? (10 points)

Loan structures and all the material to answer this question (all points) were discussed in Lectures 4 and 5. Students were also shown how to use the calculator to answer this type of queries.

The LTV implies a loan of 3mio.

For the CPM the payments are constant at 17986.52.

For the IO, payments are at 15,000 (3mio * 6%/12) except for the last period in which the balloon payment of £3mio is also due.

For the CAM, first one needs to consider the constant amortization part which is 3mio/360 periods = 8333.33. The first payment is then 8333.33 + 3mio* 6%/12=23333.33.

The payment in month 12 is (found using a recursive formula): 8333.33 + (3mio – 11*8333.33)*6%/12 = 22874.997. The payment in month 60 (end of year 5) is: 8333.33 + (3mio – 59*8333.33)*6%/12= 20874.998. Finally the payment in the last month is 8333.33 + (3mio – 359*8333.33)*6%/12 = 8375.

b- How much principle have you reimbursed for these three alternative loans after one year and after five years? How much interest have you paid cumulatively up to that point? (10 points)

The IO is very simple: the amortization will be zero while the total interest payments will 15000*12 = 180000 and 15000*60 = 900000.

For the CPM, this can be obtained from the AMORT worksheet of the calculator setting P2 to 12 and to 60. This gives total amortization in years 1 and 5 respectively at 36840.35 and 208369.30 and interest payments at 178997.84 and 870821.65.

For the CAM, students really have to work through the maths. The amortization in years 1 and 5 is simply 8333.33 * 12 = 100000 and 8333.33 *0 60 = 500,000. The interest payments are determined by the following recursive equation:

INT1= 3mio * 6%/12; INT2= (3mio – 8333.33) * 6%/12; INT3= (3mio – 8333.33*2) * 6%/12; …; INT12 = (3mio – 8333.33*11) * 6%/12. These can be regrouped as follows: 3mio * 6%/12 * 12 – 8333.33 * 6%/12 * (1+2+3+4+5+6+7+8+9+10+11) [the latter expression summing up to 66]. All in all, this gives total INT payment up to 12 = 180,000 – 2749.99 = 177250

For total payments to 60 periods (end of year five) this will be driven by the following expression:

$$Total\ INT = 3mio \times 60 \times \frac{6\%}{12} - 8333.33 \times \frac{6\%}{12} \times \sum_{n=1}^{59} n$$

The last bit of the expression is 1+2+3+4+5+6+…+58+59 which can be computed and equals 1770. Overall this gives rise to a total of 826250.03.

c- Suppose you have a monthly income of £24,000 and the bank is imposing a debt-coverage ratio of at least 1.2. Which mortgage do you qualify for? (5 points)

The person would qualify only for the IO and CPM. Not for the CAM.

d- Imagine your initial income at £24,000 is relatively low but expected to grow over time. Which loan would you prefer considering the schedule of payments? Why? (10 points)

There is no set answer. But qualifying and aligning payments and income growth/potential should be paramount in students' discussions. The CAM is the worse (no qualify), IO is good if people can "force themselves" to save. CPM is a good balance.

e- Which loan does the bank prefer and why? (10 points)

The paramount considerations for the bank are default risk and pre-payment risk. The CAM is the safest from the point of view of default, the IO the worst. Once again, there is no set answer.

f- Imagine now you are considering the CPM mortgage but would like to reduce its monthly payments to just above those of the IO payments – that is at £15,500. What would be the outstanding amount of the loan after 30 years? And what would be the outstanding amount of the loan after 30 years if you reduced the CPM payments to exactly match those of the IO loan? What do you notice? (10 points)

These are examples of negative/partial amortization found by setting payments in the worksheet and finding the future value. The answer to the first sub-question is 2497742.48. The answer to the second part is exactly 3mio. If the CPM payment is set to the IO, there is no amortization and the two loans basically coincide.

g- You are instead considering leaving £1,500,000 in outstanding loan at maturity – what would be the monthly payments in this case? (5 points)

This can be found by setting the future value to 1.5mio and treating the whole loan as one stretching for 360 periods and partially amortizing. The payment would be 16493.26.

h- Consider again the original CPM (LTV 50%; monthly compounding; 30 years maturity; 6% interest charge). The bank is applying three discount points. What is the effective cost of borrowing if you hold the mortgage to maturity? What would be the effective cost of borrowing if instead you repaid after five years, two years and one year? (10 points)

The first step is to recalculate the loan in full to get again the payments to 17986.52. The discount points are 3% of 3mio = 90000. If the actual loan of 2.91mio is held to maturity with payments of 17986.52, it gives an

IRR of 6.2865%. To work out the other IRR, one needs to find the outstanding balances at the end of year 1, year 2 and year 5. These are respectively: 2963159.65; 2924047.06 and 2791630.70. The corresponding IRRs are 9.1684%; 7.6413% and 6.7306%.

i- Comments on your findings at point h- and discuss what the rationale is for the bank to impose discount points. Why would individuals accept discount points? (10 points)

The students should comment that the sooner the repayment occurs, the higher the IRR. This is because the discount points impose a fixed cost that is less incident on the percentage cost if it's spread out over a longer period. The students should add that banks impose discount points to cover possible cost of prepayments – which are more likely to occur when rates fall and people refinance, forcing the bank to reinvest at lower rates – as well as to "positively select" non-mobile individuals. Individuals would take discount points as an option to prepay – where the alternative is a mortgage that does not allow for this – or to qualify – if alternative loans have no points but charge a higher nominal rate (to match the higher effective IRR of the loan with discount points).

j- You are now offered an adjustable rate mortgage (ARM). The structure is as follows. The loan still lasts 30 years and has monthly compounding. The initial interest rate is still 6%. This is expected to drop to 4% from the beginning of the third year to the end of the fourth, and to increase to 8% from the beginning of the 5th year till the end of the contract. The bank is still applying three discount points. What are the payments you owe to the bank in in month 1, month 25 and month 49? What is the IRR on this loan if you hold it to maturity? What is instead the IRR if you repay at the end of the fourth year? Discuss the results you obtain (20 points)

ARMs are essentially CPMs that "update" any time the interest rate resets. So the students would have to start again with the original CPM to work out that the payment is 17986.52. This goes from period 1 to 24, when the rate adjusts and the outstanding loan is 2924047.064. This is the new amount to use to calculate a loan with 4% interest over 360-24 = 336 periods – or simply 28 years. This gives payments of 14480.2418 for periods 25-48. The outstanding balance at this stage is 2805981.94 which will be charged at 8% for the rest of the loan's life – i.e. 26 years or 312 periods. This gives a payment of 21398.3654.

With 3 discount points the IRR to maturity is 7.3069%. This can be worked out with the cash flow worksheet with a loan of 97% of 3mio vis-à-vis payments of 17986.52 for 24 periods, 14480.2418 for 24 periods and 21398.3654 for 312 periods. If the loan instead is repaid after 48 periods, the students have to consider 24 periods at 17986.52, 23 periods at 14480.2418 and add the outstanding balance to the 48th period. This gives an IRR of 5.9432%.

This is below the one at maturity – which is counterintuitive as discount points should lead to a higher rate the earlier I repay. However in this case by repaying the borrower is avoiding a long period charged at 8%. I expect the students to be able to see this logic and comment on it.

Question 3

You are considering purchasing the following building as a real estate investment with a horizon of four years. The structure has three floors and each floor comprises of 15,000 square metres. On each floor, 1000 square metres (out of the 15000) are occupied by restrooms, lobby areas, lifts and other services. They are therefore "unusable" from the point of view of the activities of your planned tenants. At the moment, the bottom floor has two tenants – a consultancy and a marketing company – each occupying half of the space. Their leases have already been signed and will expire after you are planning to sell the building. These are worth £300 per square metre per year. Both tenants will further pass an overage rent of 10% of their revenues starting from the third year after you have bought the property. Their revenues are £1,500,000 (for the marketing company) and £3,000,000 (for the consultancy). Rents for these two tenants are charged on the usable space they occupy while the service area space is apportioned in equal shares and charged at £100 per square metre per year. The first floor of the building is instead all occupied by one tenant – a financial firm. The lease has been already agreed at £350 per square metre per year. The company got several concessions. First, they are not paying any rents on the service areas. Second, the company benefits from a rent discount of 20% for the first two years of your investment. Last, the first rent review will not occur till the year after you have sold the building – when the lease is expected by increase by 10%. The top floor is instead vacant and in need of redevelopment. Your strategy for this space is as follows. You will spend 6 months renovating all the space at a cost of £50 per square metre. You will then spend a further £750,000 in marketing of the space, brokerage and other fees to let out the space (for simplicity assume that these costs are paid out at the end of the first year and ignore within year discounting). You set a high rent per square metre per year – at £400 – but include a 25% concession for the first 1.5 years from the end of the renovation works. You further expect a rent review at the beginning of the fourth year pushing up the rent by 10%. Since the whole space is being renovated and will be "bespoke", you are planning to charge the tenant for the whole floor, including service areas. You expect a 50% chance that the space will be vacant for the first 6 months following renovation, but occupied after that.

a- Put together the operation pro-forma of this investment that will constitute the basis of your discounted cash flow (DCF) analysis. Then estimate the resale price considering you expect the relevant exit yield to be 8%. (20 points)

All the elements needed to answer this question were explained during Lectures 1, 2 and 3 – except for the CPM mortgage calculations which were discussed in Lecture 4.

The table for the DCF should be assembled as follows. Note that time 5 is needed to calculate the resale value at time 4 using a cap rate of 8%.

	Time 1	Time 2	Time 3	Time 4	Time 5
Ground floor – Tenant 1	7000m *£300 + 500m * £100 = 2.15mio	7000m *£300 + 500m * £100 = 2.15mio	7000m *£300 + 500m * £100 + 1.5mio * 10%= 2.30mio	7000m *£300 + 500m * £100 + 1.5mio * 10%= 2.30mio	7000m *£300 + 500m * £100 + 1.5mio * 10%= 2.30mio

Ground floor – Tenant 2	7000m *£300 + 500m * £100 = 2.15mio	7000m *£300 + 500m * £100 = 2.15mio	7000m *£300 + 500m * £100 + 3mio * 10%= 2.45mio	7000m *£300 + 500m * £100 + 3mio * 10%= 2.45mio	7000m *£300 + 500m * £100 + 3mio * 10%= 2.45mio
First floor	14000m * £350 * (1-20%) = 3.92mio	14000m * £350 * (1-20%) = 3.92mio	14000m * £350 = 4.9mio	14000m * £350 = 4.9mio	14000m * £350 (1+10%) = 5.39mio
Top floor	Costs: £50 * 15000m + 750000 = 1500000 Revenues: £400 * ½ year * 1-50% vacancy * 1-25% discount * 15000m = 1125000 Total = -375000	£400 * 1-25% discount * 15000m = 4.5mio	£400 * 15000m = 6mio	£400 * (1+10% review) * 15000m = 6.6mio	£400 * (1+10% review) * 15000m = 6.6mio
Grand total	7.845mio	12.720mio	15.650mio	16.250mio	16.740mio

The sale price given the total NOI in period 5 is 209.25mio.

I always recommend the students to write intermediate steps and how they interpreted my assumptions so that I can check where calculations went wrong – if the numbers don't like up.

b- If you are asked to pay £180,000,000 for the building, what would be the IRR of this investment? (5 points)

This gives an IRR of 10.561%

c- Imagine now you leverage your investment. Given the short horizon, you take an Interest-Only (IO) loan with an LTV of 70% over four years and charging 7% per year with annual compounding. What is the IRR with gearing? Was this finding expected? (10 points)

With leverage at 70% LTV (126mio of 180mio) and charging 7% per year, the IRR becomes 17.375%. The finding is expected because the IRR before leverage is above the cost of borrowing.

d- What would have been the IRR of the leveraged investment had you worked this out using a Weighted-Average Cost of Capital (WACC) formula. Discuss your results relative to what you found at point c-. (10 points)

The WACC would have given 10.561% + (10.561% - 7%) * 7/3 = 18.87%. The number is not equal to the one obtained in c-. This is not surprising: WACC is an approximation holding D and E constant – but here E increases (because of asset appreciation) so the WACC overstates returns.

e- Given this loan structure, calculate the cash-on-cash yield in the first and fourth year, and the annualized growth yield of your investment. What do you notice? (10 points)

The cash-on-cash for year 1 is 4.3583% - 70% * 7% / 1-70% = -1.8056%. The cash-on-cash yield for year 4 9.0278% - 7% * 70% / 1-70% = 13.7593%. The annualised growth yield can be found by comparing equity at

the beginning and at the end: 54mio and 83.250mio. To find the constant growth are leading from 54 to 83.250, students need to solve for X in the following equation: 54 (1+x)^4 = 83.250 → x = (83.250/54)^1/4 -1 = 11.4288% per year.

This investment therefore has a poor cash-on-cash yield at the beginning because of redevelopment and marketing costs but then it becomes significantly profitable. Notice that the negative cash-on-cash corresponds to a negative NOI after mortgage payments in year 1 – so students might want to discuss whether this is allowed or some equity cure might need to be injected.

f- Why is it useful from a "risk assessment" point of view to investigate the variation of cash-on-cash yields over the course of the investment and the overall growth yield of your asset? (5 points)

The usefulness of these concepts is that they allow an investor to better understand the risk/return profile of an investment.

g- Consider now an alternative loan: a fully-amortizing constant payment mortgage (CPM) over 25 years. The loan still has an LTV of 70% and charges 7% per year. Compounding is done on a yearly basis. You still plan to hold your investment for four years and reimburse the loan at the time when you sell the building. What is the IRR of your investment with this alternative gearing? Discuss your results and compare them to those presented at point c-. (15 points)

The CPM would give a yearly payment of 10812125.17. The outstanding balance in year 4 is 117155077.8. The IRR on this investment is 16.9606%. This is lower than with the IO for two reasons: the first period sees a large negative NOI due to large mortgage payments. This has a large effect on the IRR because period 1 is not heavily discounted. Conversely this loan gives a higher resale value because of amortization. While good, this is not very consequential because it's heavily discounted as it occurs in period 4.

h- What answer would have you obtained using a WACC approach? What do you notice? (5 points)

The WACC would have given the same IRR as for the IO loan further neglecting the fact that here both E and D change over time because of asset appreciation and amortization.

i- Consider the following alternative strategy for your top floor (the plans for the other two floors are unchanged). You still forecast to have the same costs and plans for redevelopments as well as for brokerage, marketing and service area pricing. However, you are planning to market the property at £300 per square metre per year. You also expect to give a lower concession of 10% (still for 1.5 years from the end of the renovation works) and still have a rent review at the beginning of year four pushing up the charge per square metre per year by 10%. Given the lower rent, you expect the probability of vacancy in the first six months to drop to 25% (and to have fully rented your space thereafter). What is the IRR of this investment? How does it compare to the one obtained following alternative strategy discussed in the main instructions of this question? Discuss the main forces driving your results. And what would be the IRR if you took up the IO loan? What do you notice? (20 points)

The various quantities for the ground floor and for the first floor remained unchanged. So if a detailed DCF has been assembled, they can be easily be copied over. On the other hand, the calculations for the top floor are now as follows:

Top floor	Costs: £50 * 15000m + 750000 = 1500000 Revenues: £300 * ½ year * 1-25% vacancy * 1-10% discount * 15000m = 1518750 Total = 18750	£300 * 1- 10% discount * 15000m = 4.050mio	£300 * 15000m = 4.5mio	£300 * (1+10% review) * 15000m = 4.95mio	£300 * (1+10% review) * 15000m = 4.95mio
Grand total	8.238750mio	12.270mio	14.150mio	14.600mio	15.090mio

This leads to a sale price of 188.625mio and if the price is still 180mio, the IRR becomes 7.7975%. With IO leverage, this would be 9.4931%. So there is still some return in this asset and this strategy – and leverage is still effective. However, this approach is not as remunerative because the rent is set too low – which means that at review and when concessions expire, the rent level will be subdued. On the other hand, when leverage is taken, this approach gives a lower negative NOI – though it does not fully address the potential need for an equity cure.

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